

CO1-AT1

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1. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is randomly twice as likely to be selected as each of the other sections due to higher usage. A selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?

Answer:

Prior probabilities

$$P_A = 1/4$$

$$P_B = 1/2$$

$$P_C = 1/4$$

Probability of selecting a blue car from each section

$$P_{\text{Blue}_A} = 3/10$$

$$P_{\text{Blue}_B} = 6/15$$

$$P_{\text{Blue}_C} = 2/15$$

Total probability of selecting a blue car

$$P_{\text{Blue}} = (P_A * P_{\text{Blue}_A}) + (P_B * P_{\text{Blue}_B}) + (P_C * P_{\text{Blue}_C})$$

Bayes theorem

$$P_{C_{\text{given_Blue}}} = (P_C * P_{\text{Blue}_C}) / P_{\text{Blue}}$$

```
print("Probability that the blue car came from Section C =", round(P_C_given_Blue, 4))
```

```
Probability that the blue car came from Section C = 0.1081
```

2. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

1. (a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4

2. (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2

3. (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
4. (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
5. (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
6. (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
7. (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
8. (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2
9. i) Identify the epoch at which the model begins to overfit.
10. ii) Suggest two effective techniques to reduce overfitting in this model.
11. iii) Discuss how overfitting affects the model's generalization ability on unseen data.

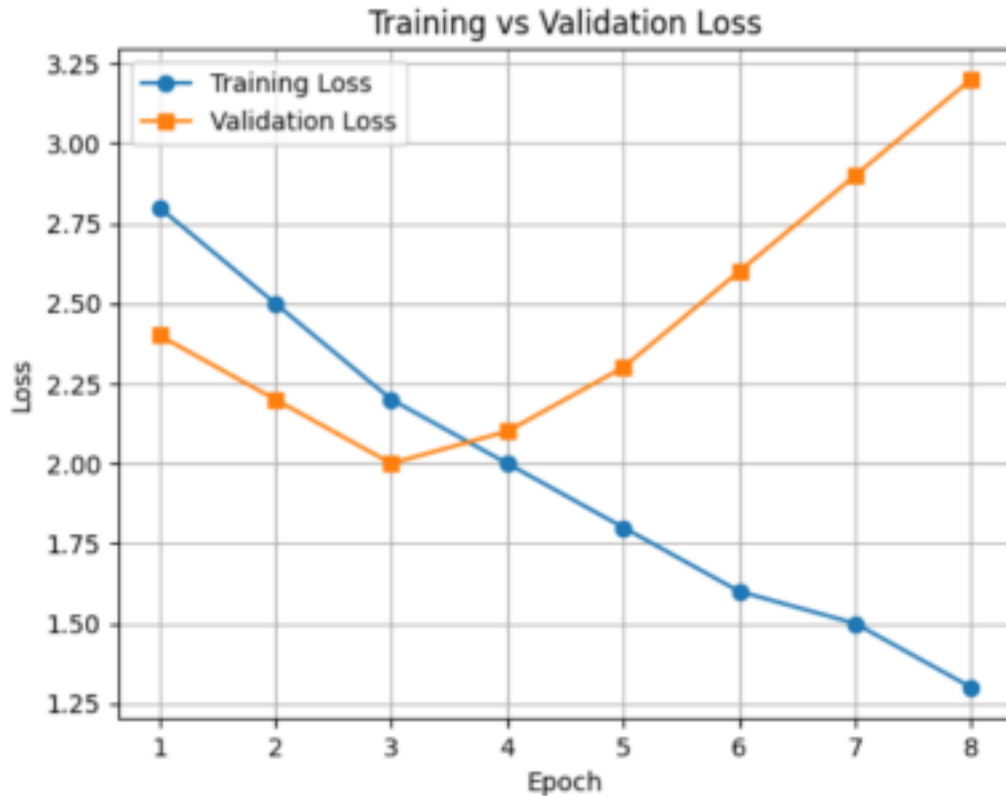
Answer:

```
import matplotlib.pyplot as plt

epochs = [1,2,3,4,5,6,7,8]
training_loss = [2.8,2.5,2.2,2.0,1.8,1.6,1.5,1.3]
validation_loss = [2.4,2.2,2.0,2.1,2.3,2.6,2.9,3.2]

plt.plot(epochs, training_loss, marker='o', label="Training Loss")
plt.plot(epochs, validation_loss, marker='s', label="Validation Loss")
plt.title("Training vs Validation Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()
plt.grid(True)
plt.show()

# Find overfitting point
for i in range(1, len(validation_loss)):
    if validation_loss[i] > validation_loss[i-1]:
        print("Model starts overfitting at Epoch", epochs[i])
        break
print("\nTechniques to reduce overfitting:")
print("1. Early Stopping")
print("2. Dropout / Regularization")
print("\nEffect of Overfitting:")
print("The model memorizes training data and performs poorly on unseen data.")
```



3.Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

12. (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%

13. (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%

14. (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%

15. (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%

16. (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%

17. (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%

18. (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%

19. (h)Epoch 8: Training Accuracy = 92%,Validation Accuracy = 68%

20. i) Identify the epoch at which the model starts overfitting.

21. ii) Explain why validation accuracy decreases despite training

accuracy increasing. 22. iii) Suggest two methods to improve

validation performance.

Answer:

```
import matplotlib.pyplot as plt
```

```
epochs = [1,2,3,4,5,6,7,8]
```

```
training_accuracy = [60,65,70,75,80,85,88,92]
```

```
validation_accuracy = [58,63,68,72,73,72,70,68]
```

```
plt.plot(epochs, training_accuracy, marker='o', label="Training Accuracy")
```

```

plt.plot(epochs, validation_accuracy, marker='s', label="Validation Accuracy")

plt.title("Training vs Validation Accuracy")

plt.xlabel("Epoch")

plt.ylabel("Accuracy (%)")

plt.legend()

plt.grid(True)

plt.show()

# Find overfitting point
for i in range(1, len(validation_accuracy)):
    if validation_accuracy[i] < validation_accuracy[i-1]:
        print("Model starts overfitting at Epoch", epochs[i])
        break

print("\nReason for decrease in validation accuracy:")

print("The model learns the training data too closely and fails to
generalize.") print("\nMethods to improve validation performance:")

print("1. Early Stopping")

print("2. Data Augmentation / Regularization")

```

